Software Requirements Specification (SRS) Template

Project: Online Exam Portal
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Group Number : 06

AUTHORS

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Revision history

Version	Date	Author	Change summary	Approval
1.0	13-09-2026	Team 6	Initial SRS draft covering introduction, scope, functional/non-functional requirements, security section, use-case diagrams, and RTM for the Online Exam Portal	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for the Online Exam Portal, a web-based system for conducting MCQ-based examinations with automated evaluation.

1.2 Scope

Covers student registration/login, exam creation, MCQ test attempts, autoevaluation, result reporting, and admin monitoring. Excludes proctoring/webcam invigilation and integration with external LMS platforms.

1.3 Audience

Developers, QA Engineers, Instructors/Admins, Students, and Assessment Evaluators.

1.4 Definitions

List of acronyms: MCQ, UI, API, DB, JWT, HTTPS, LMS.

2. Overall description

2.1 Product perspective

The Online Exam Portal is a web application with separate student and instructor/admin interfaces, a backend API, and a database. It supports concurrent exam attempts, server-side timing, and automatic grading.

2.2 Major product functions (detailed)

- Authenticate user (student/instructor)
- Create and configure MCQ exams
- Attempt exam with countdown timer
- Auto-evaluate and score submitted answers
- Display results and performance analytics
- Schedule exams and manage question bank
- Admin dashboard for monitoring attempts
- Basic auto-save of in-progress answers on connectivity loss

2.3 User roles and characteristics (expanded)

- Student: basic computer literacy, expects a clear, distraction-free testtaking UI.
- Instructor/Admin: creates exams and question banks, reviews analytics. - Evaluator: reviews auto-graded results and flags disputes.

2.4 Operating environment

Modern web browser (Chrome/Firefox/Edge) on desktop or mobile, Node.js/Express or Django backend, MongoDB/PostgreSQL database, hosted on cloud or campus server.

2.5 Constraints

Server-side timer enforcement to prevent client-side tampering, browser compatibility, concurrent-user load during scheduled exam windows.

3. External interface requirements

3.1 User interfaces

Primary UI: responsive web UI with a clean test-taking screen, question navigator, and timer. Accessibility: screen-reader friendly labels and highcontrast mode.

3.2 Hardware interfaces

- Standard keyboard and mouse/touchscreen
- Webcam (optional, future proctoring)
- Any HTML5-compatible display device
- No specialised peripherals required
- Printer (optional, for result/receipt export)
- Admin monitoring console (browser-based)

3.3 Software interfaces

- Backend REST API (JSON over HTTPS) for auth, exam data, and submissions.
- Database service (MongoDB/PostgreSQL) for question bank, attempts, and results.
- JWT-based authentication service for session management.

3.4 Communications

- HTTPS enforced for all client-server communication.
- Client-side auto-save with retry/backoff if connectivity drops midexam.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow EXAM-F-###.

4.1 Authentication

Description: Authenticate students and instructors via registered credentials before granting access to exams or the admin panel.

Req ID	Requirement	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
EXAM- F-001	The system shall authenticate users via email/username and password before allowing access.	Functional	High	Security / Academics	ACEXAM-F-001: Valid credentials return a JWT session token. Test: TC-Auth-01	Requires user DB and JWT service

EXAM- F-002	The system shall hash and salt passwords; passwords are never stored or logged in plaintext.	Functional	High	Security	ACEXAM-F-002: Password hash verified in DB; absent from logs. Test: TC-Auth-02	Audit review required
EXAM- F-003	The system shall lock the account after 5 consecutive failed login attempts and log the event.	Functional	High	Security	ACEXAM-F-003: After 5 failures, further attempts are blocked and a message displayed. Test: TC-Auth-03	Account flagged for admin review

4.2 Exam Attempt & Timer

Description: Allow a student to attempt an MCQ exam within a fixed, serverenforced time limit.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies

EXAMF-010	The system shall load exam questions in randomized order and start a serverside countdown timer.	Functional	High	Business	ACEXAM-F-010: Timer starts on exam load and matches configured duration. Test: TC-Exam-01	Requires question bank service
EXAMF-011	The system shall autosubmit the exam when the timer expires, saving all answered questions.	Functional	High	Operations	ACEXAM-F-011: Submission recorded at timer expiry with no data loss. Test: TC-Exam-02	Requires reliable clientserver sync

4.3 Question Bank Management

Requirements: The system shall let instructors create, edit, tag by topic/difficulty, and delete MCQ questions with a single correct answer and 3 distractors; support bulk import via CSV; and version question banks per exam. IDs: EXAM-F020..EXAM-F-025.

4.4 Evaluation & Results

Requirements: The system shall auto-grade MCQ submissions on completion, compute scores and percentile, and display a results/analytics dashboard to students and instructors. IDs EXAM-F-030..EXAM-F-034.

4.5 Admin Dashboard & Monitoring

Requirements: Provide an admin dashboard to schedule exams, monitor live attempt counts, view audit logs, and export result reports. IDs EXAMF-040..EXAMF-044.

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs EXAM-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
EXAM-NF-001	Exam page load and answersubmission response shall be ≤ 2 seconds for 90% of requests under normal load.	Performance	High	90th percentile ≤ 2s in productionlike test. Test: TC-Perf-01
EXAM-NF-002	System shall provide 99.5% availability during scheduled exam windows.	Reliability	High	Uptime reports show ≥99.5% during exam windows. Test: Ops reports.
EXAM-NF-003	Passwords and session tokens must be stored securely; passwords are never stored in plaintext.	Security/ Compliance	High	Security audit checklist pass. Test: TC-Sec-01

EXAM-NF-004	System shall log all exam attempts and submissions with timestamps, retained for 1 academic year.	Audit/Data Retention	High	Logs retrievable per attempt ID for audit. Test: TC-OPS-01
EXAM-NF-005	System shall support up to 200 concurrent exam attempts without degraded response time.	Scalability	Medium	Load test with 200 concurrent users passes. Test: TC-Load-01

5.1. Security

5.1.1 Security Objectives

1) Protect student credentials and session tokens from unauthorized access. 2) Ensure exam questions and answer keys cannot be accessed or tampered with before or during an exam.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
EXAM-SR-001	The system shall enforce HTTPS/TLS 1.2+ for all client-server communication.	Security	High	
EXAM-SR-002	The system shall hash and salt all stored passwords using a strong algorithm (e.g. bcrypt).	Security	High	
EXAM-SR-003	The system shall restrict question bank and answerkey access to authorised instructor/admin roles only.	Security	High	

EXAM-SR-004	The system shall use signed, expiring JWT tokens for session management and reject tampered tokens.	Security	High	
EXAM-SR-005	The system shall log all admin actions (question edits, exam scheduling) for audit purposes.	Security	Medium	

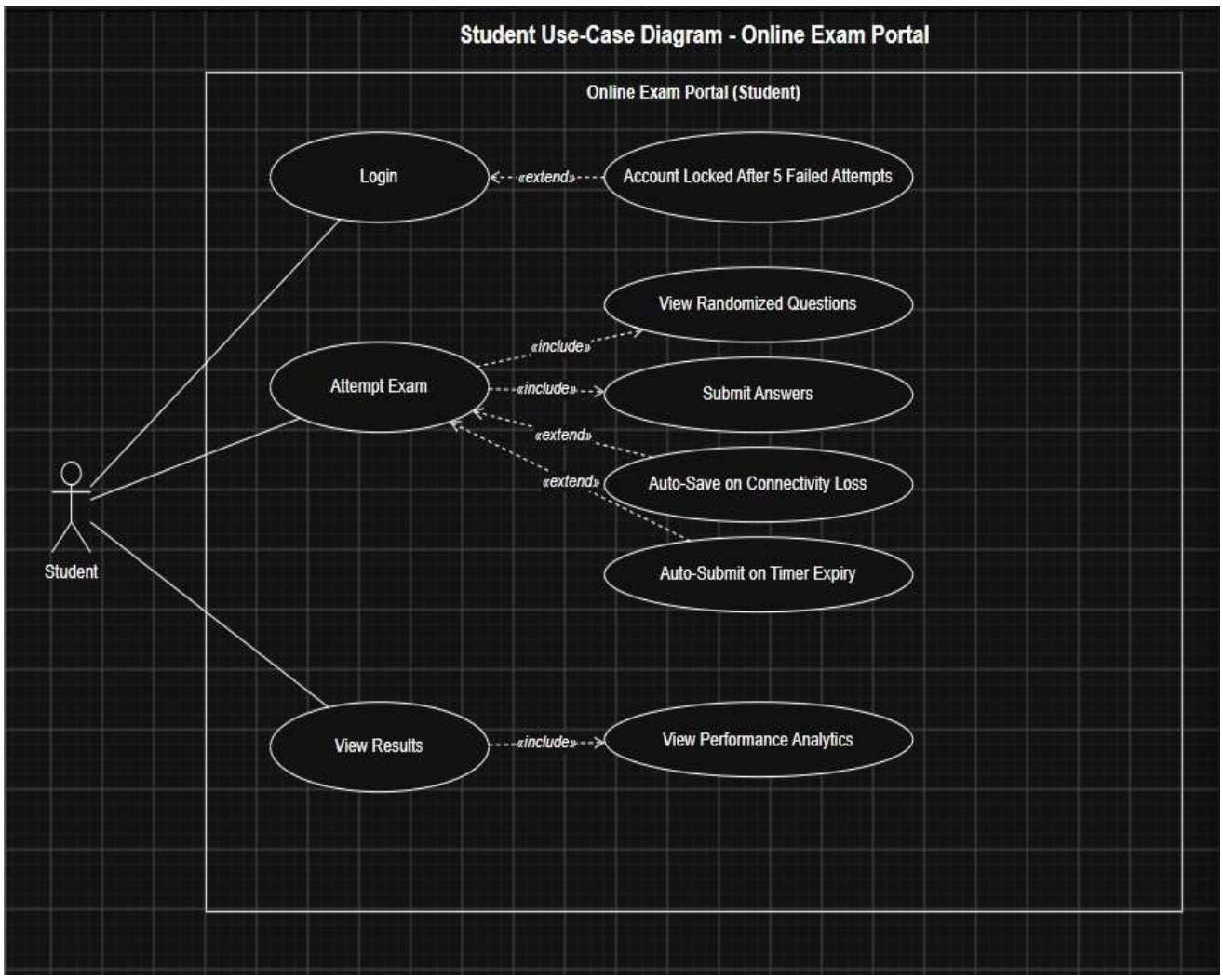
6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR failures, and RTM shows all test cases passed.
- Acceptance test suites: Authentication, Exam Attempt, Question Bank, Evaluation, Performance, Security, and Load tests.

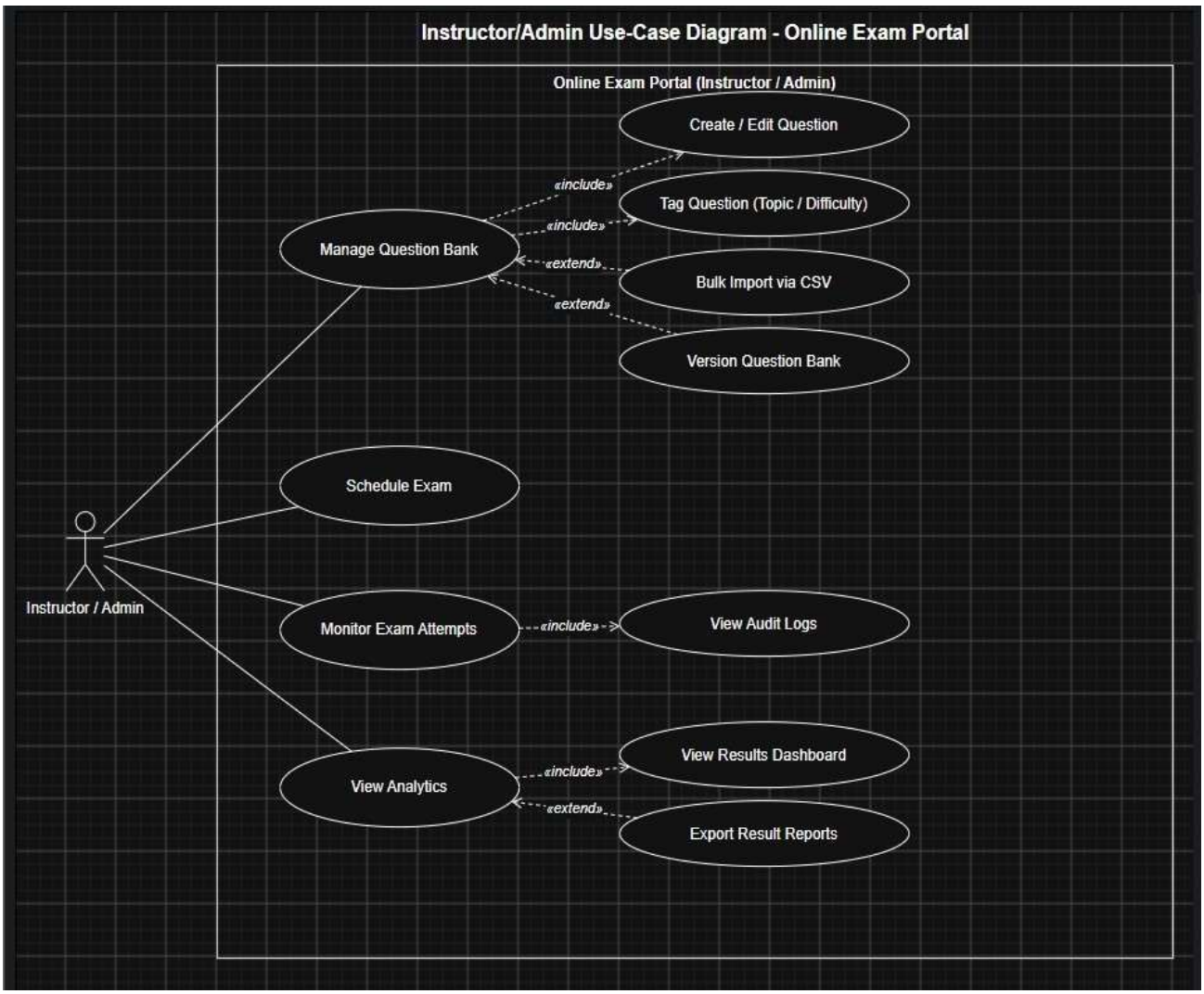
7. System models and diagrams

7.1 UML Use-Case diagram

(1) Student use cases - login, attempt exam, view results;



(2) Instructor/Admin use cases - manage question bank, schedule exam, view analytics.



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
EXAMF-001	Authenticate user	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	
EXAMF-010	Attempt exam & timer	4.2 / DS-Exam-01	ExamModule	TCExam-01 , TC-Exam-02	N	
EXAM-NF-001	Response time target	5 / DS-Perf-01	WebUI / API	TC-Perf-01	N	